

Institución: Universidad Politecnica de Yucatan

Materia: Programacion (2° Cuatrimestre)

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Selective and Iterative Control Structures

Problem 1 — Grade Averaging System The user enters student grades one by one. The program keeps reading grades until the user types "done". Then it calculates the average and displays whether the student passed (≥ 7.0) or failed. If no grades were entered, display an error message.

```
# INPUT
grades = []
while True:
    user_input = input("Enter grade or 'done': ")
    if user_input.lower() == 'done':
        break
    # PROCESS
    grades.append(float(user_input))

# PROCESS & OUTPUT
if len(grades) == 0:
    print("No grades entered. Please enter at least one grade.")
else:
    average = sum(grades) / len(grades)
    # OUTPUT
    print(f"Average: {average:.2f} — {'Passed' if average >= 7.0 else 'Failed'}")
```

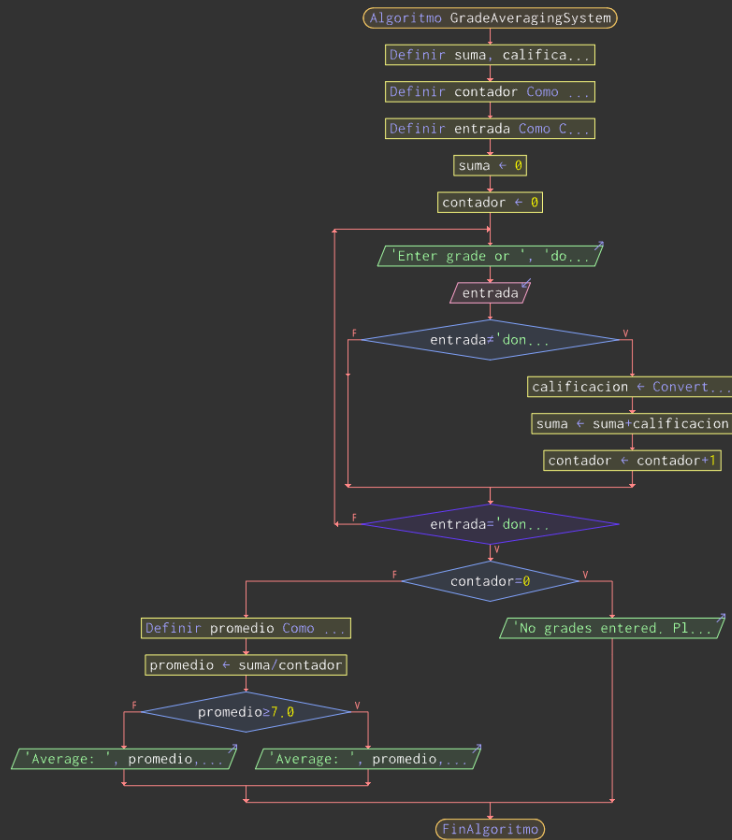


Figura 1: FlowChart_Ejercicio 1

```

Enter grade or 'done': 8
Enter grade or 'done': 9
Enter grade or 'done': 10
Enter grade or 'done': 8
Enter grade or 'done': done
Average: 8.75 — Passed
  
```

Figura 2: Resultado

Problem 2 — BMI Calculator The user enters weight (kg) and height (m) for multiple people. The program calculates the BMI and classifies it for each person. The loop continues until the user types "exit" when prompted for weight. BMI formula: $BMI = \text{weight} / \text{height}^2$ Categories: Underweight (< 18.5) / Normal ($18.5-24.9$) / Overweight ($25-29.9$) / Obese (≥ 30)

```
# INPUT / PROCESS
while True:
    weight_input = input("Enter weight (kg) or 'exit': ")
    if weight_input.lower() == 'exit':
        break
    weight = float(weight_input)
    height = float(input("Enter height (m): "))

    # PROCESS
    bmi = weight / (height ** 2)

    if bmi < 18.5:
        category = "Underweight"
    elif bmi <= 24.9:
        category = "Normal"
    elif bmi <= 29.9:
        category = "Overweight"
    else:
        category = "Obese"

    # OUTPUT
    print(f"BMI: {bmi:.2f} - {category}")
```

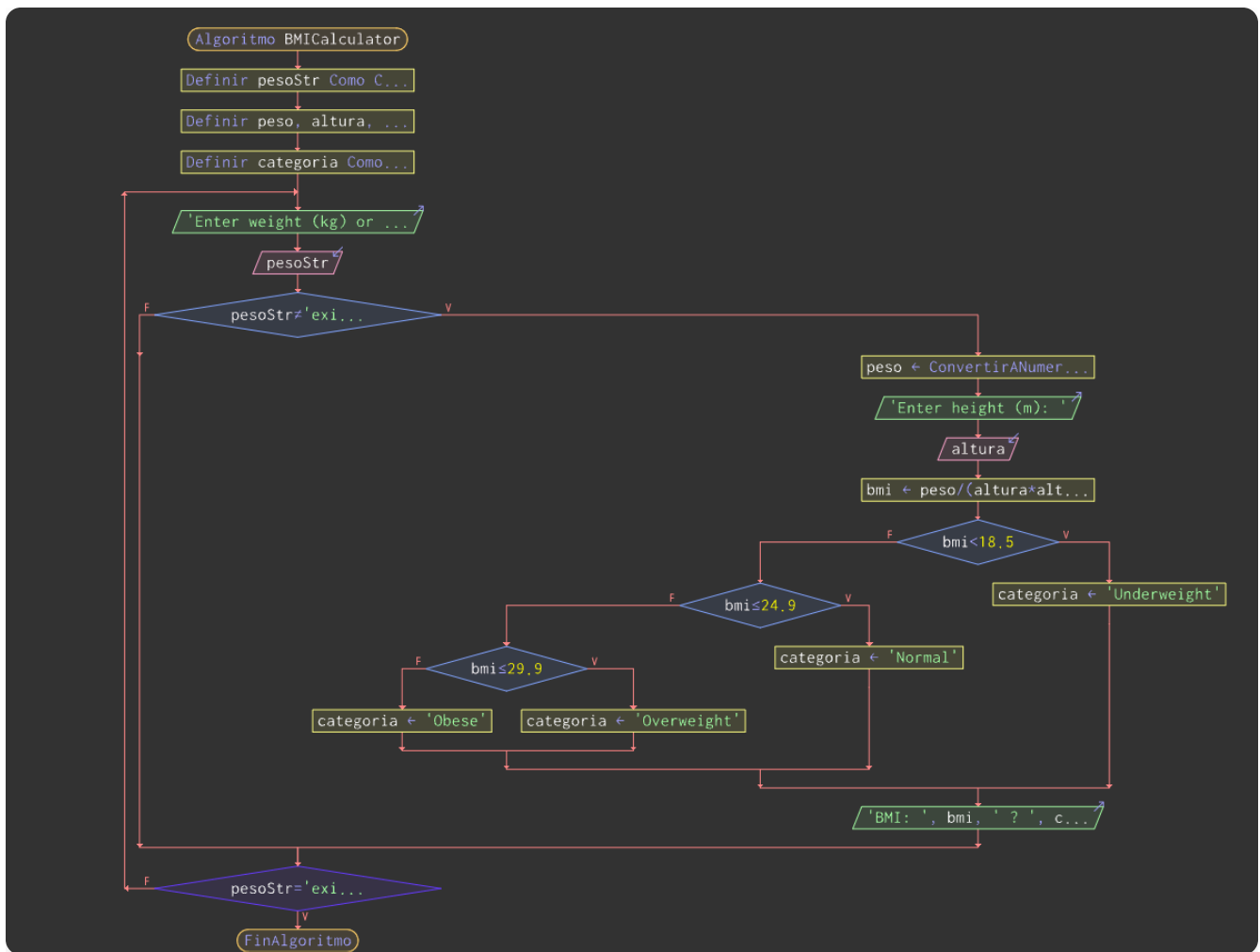


Figura 3: FlowChart_Ejercicio2

```

Enter weight (kg) or 'exit': 69
Enter height (m): 165
BMI: 0.00 – Underweight
Enter weight (kg) or 'exit': 85
Enter height (m): 1.70
BMI: 29.41 – Overweight
Enter weight (kg) or 'exit':
  
```

Figura 4: Resultado

Problem 3 – Water Bill Calculator The user enters the m³ consumed for multiple months. The program calculates the charge for each month using tiered rates. The loop continues until the

user types "exit". At the end, display the total accumulated charge.

```
# INPUT
total_accumulated = 0
while True:
    consumption_input = input("Enter consumed m3 or 'exit': ")
    if consumption_input.lower() == 'exit':
        break

    # PROCESS
    m3 = float(consumption_input)
    if m3 <= 10:
        charge = m3 * 8.00
    elif m3 <= 20:
        charge = m3 * 12.00
    else:
        charge = m3 * 18.00

    total_accumulated += charge

    # OUTPUT
    print(f"Month charge: ${charge:.2f} MXN – Total: ${total_accumulated:.2f} MXN")
```

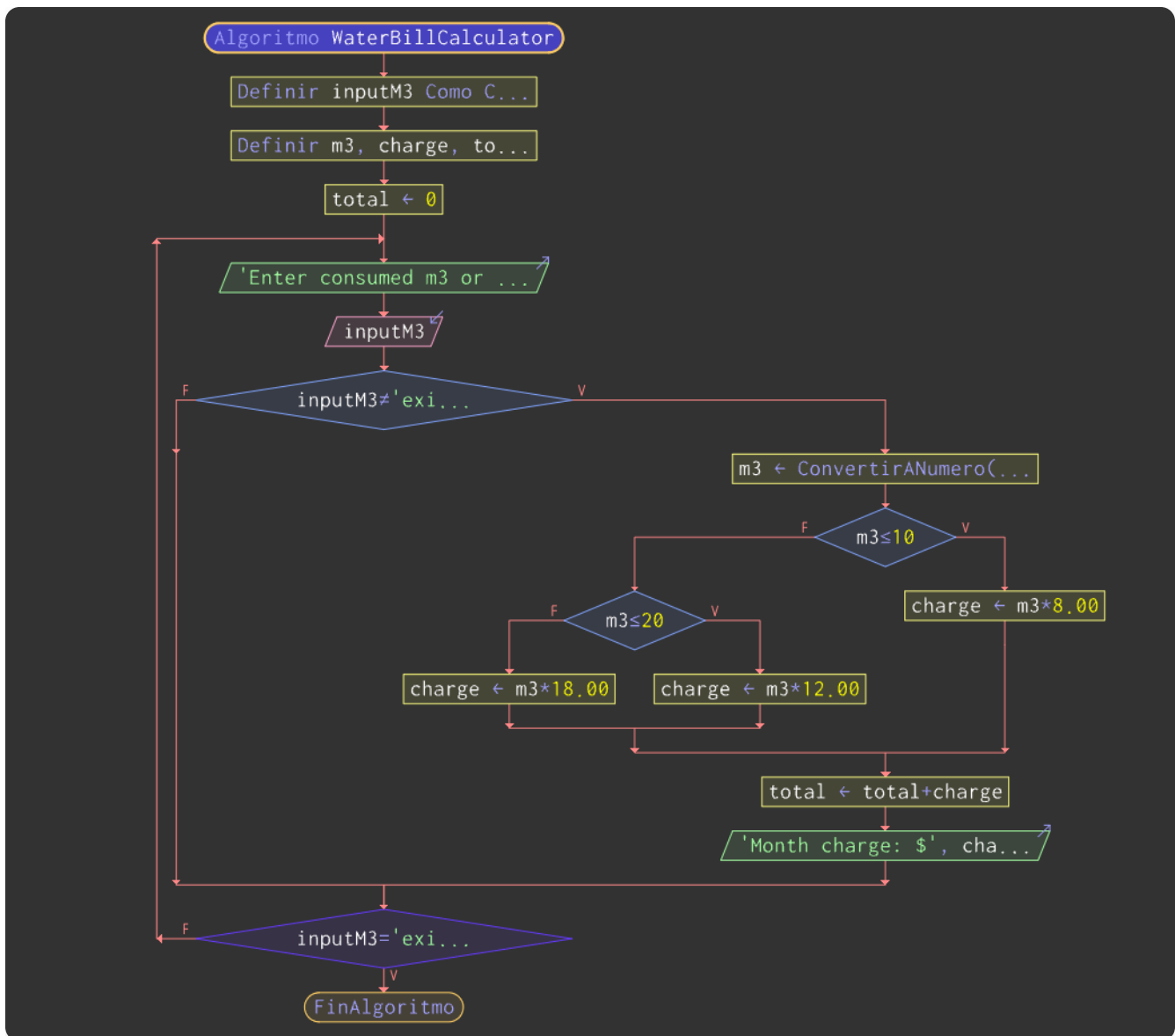


Figura 5: FlowChart_Ejercicio 3

```

Enter consumed m3 or 'exit': 15
Month charge: $180.00 MXN — Total: $180.00 MXN
Enter consumed m3 or 'exit': 15
Month charge: $180.00 MXN — Total: $360.00 MXN

```

Figura 6: Resultado

Problem 4 — Season Classifier The user enters month numbers one by one. The program displays the corresponding season for each entry. The loop continues until the user types

"exit". If a number is outside 1–12, display an error message and continue the loop.

```
# INPUT / PROCESS
while True:
    month_input = input("Enter month number (1-12) or 'exit': ")
    if month_input.lower() == 'exit':
        break

    # PROCESS
    month = int(month_input)
    if month < 1 or month > 12:
        print("Invalid month. Please enter a number between 1 and 12.")
        continue

    if month in [12, 1, 2]:
        season = "Winter"
    elif month in [3, 4, 5]:
        season = "Spring"
    elif month in [6, 7, 8]:
        season = "Summer"
    else:
        season = "Fall"

    # OUTPUT
    print(season)
```

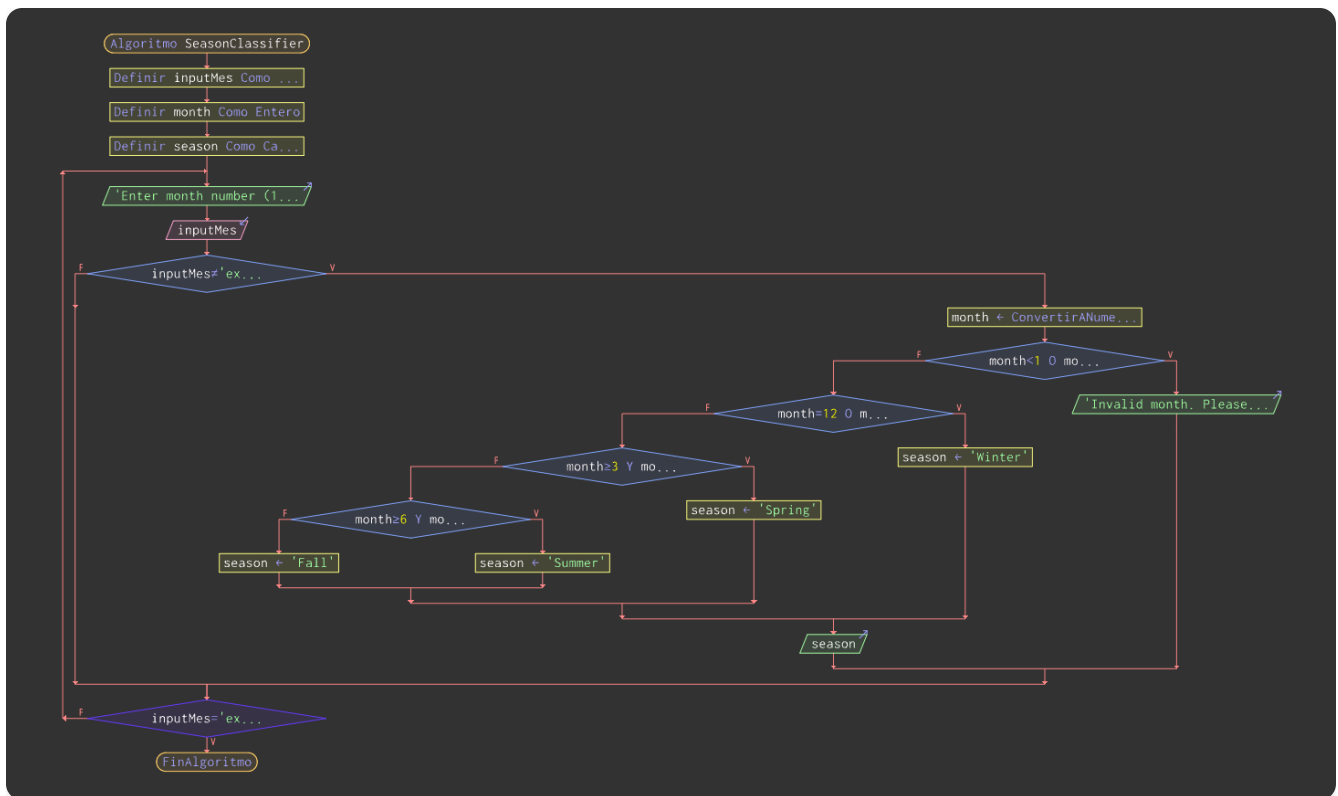


Figura 7: FlowChart_Ejercicio 4

```

Enter month number (1-12) or 'exit': 10
Fall
Enter month number (1-12) or 'exit': 5
Spring
Enter month number (1-12) or 'exit': |
  
```

Figura 8: Resultado

Problem 5 — Shipping Cost Calculator The user enters the weight (kg) and distance (km) for multiple packages. The program calculates the shipping cost for each package. The loop continues until the user types "exit". At the end, display the total accumulated shipping cost.


```
# INPUT
total_shipping = 0
while True:
    weight_input = input("Enter package weight (kg) or 'exit': ")
    if weight_input.lower() == 'exit':
        break

    weight = float(weight_input)
    distance = float(input("Enter distance (km): "))

    # PROCESS
    if distance <= 100:
        if weight <= 5:
            cost = 50.00
        else:
            cost = 80.00
    else:
        if weight <= 5:
            cost = 120.00
        else:
            cost = 200.00

    total_shipping += cost

    # OUTPUT
    print(f"Shipping cost: ${cost:.2f} MXN - Total: ${total_shipping:.2f} MXN")
```

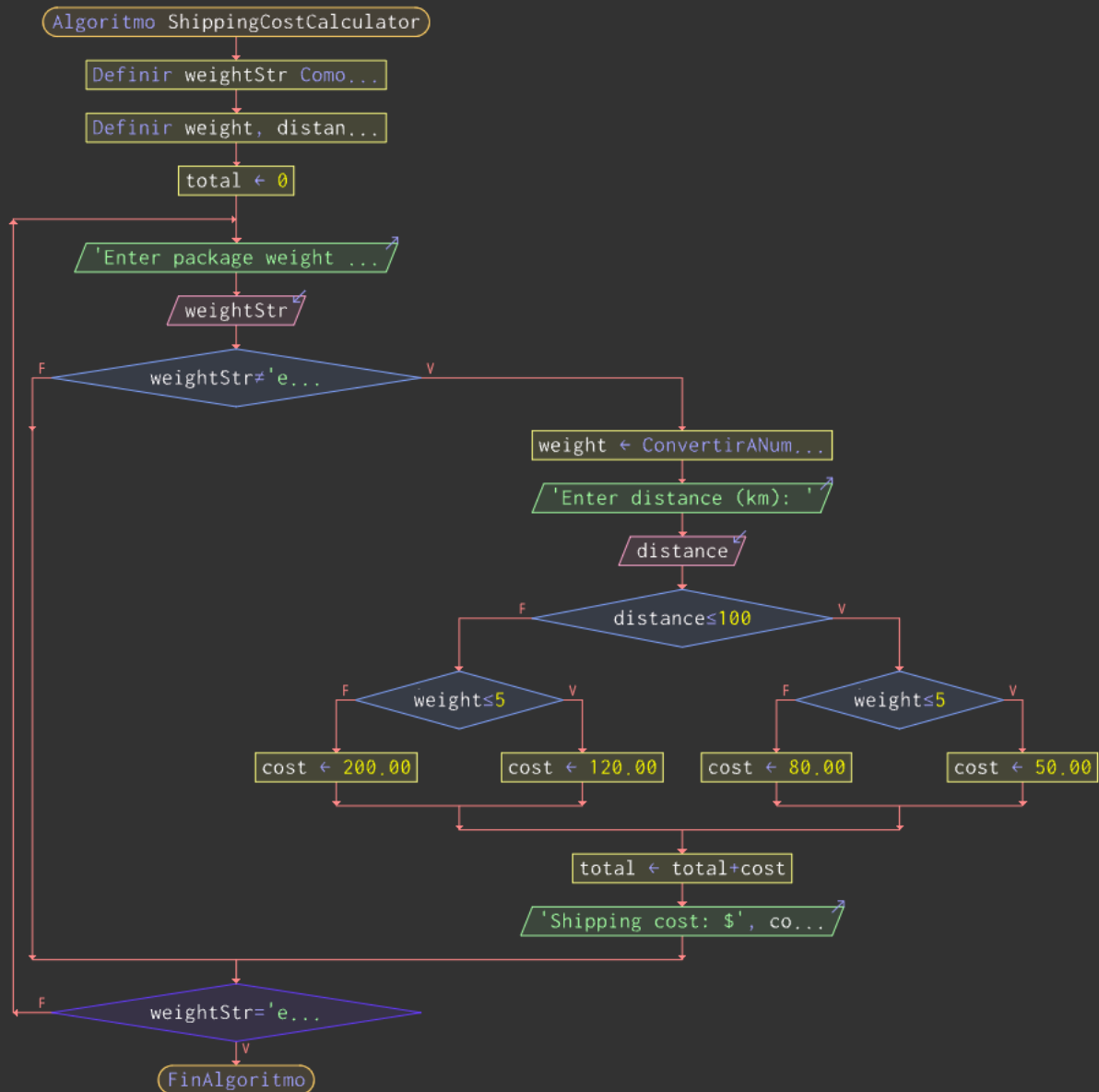


Figura 9: FlowChart_Ejercicio 5

```

Enter package weight (kg) or 'exit': 180
Enter distance (km): 1800
Shipping cost: $200.00 MXN – Total: $200.00 MXN
Enter package weight (kg) or 'exit': 250
Enter distance (km): 25000
Shipping cost: $200.00 MXN – Total: $400.00 MXN
Enter package weight (kg) or 'exit': |
  
```

Figura 10: Resultado